

Predicting House Prices Using Regression

Stat 490 Fall 2022

Alex Caramanico, John Park, Aaron Philpott

I - Background and Introduction

One of the most pertinent topics in the United States today is the price of housing. These prices have been generally increasing since the 1970s, but it may still be difficult to predict future prices due to the existence of many potentially significant variables. Hence, determining the most significant factors affecting house prices can be a daunting task. That said, it is certainly an important task because housing is one of the most basic needs for all people, thus the problem at least affects everyone in the country, and probably those outside of the United States through association. This report will seek to predict house prices by applying several regression algorithms; linear regression, Bayesian ridge regression, and gradient boost regression. These algorithms will be applied to determine the most important factors, from which point predictions can be made for new housing units. The report will also assess the effectiveness of each algorithm. Detailed descriptions of each algorithm and their assessment methods are included in a later section.

Perhaps the greatest difficulty in modeling house prices is accounting for factors which are not easily quantified. These may include location of the house and the current economic state of the country. The former could be represented by longitude and latitude, or indicator variables that represent city or proximity to various landmarks. While this may complicate result interpretation, the inclusion of such variables would give a very thorough knowledge of the unit in question. Alternatively, as in the case of the selected dataset, location can be represented by a variable corresponding to the neighborhood the house is in. Economic state, while perhaps more quantifiable, is challenging in that it can be represented by a number of different metrics. Such metrics could be GDP per capita, unemployment rate, interest rate, or others. That said, these metrics are likely to be relatively constant for each data point because they are highly dependent upon time. If the entire sample is taken within a short period of time, it is not likely for these metrics to change, at least not in a meaningful way.

These issues will be compensated for via the selected dataset, which narrows the scope to Ames, Iowa. The dataset includes a variable for the neighborhood the house is located within, allowing for analysis on the significance of location. While this may provide an elementary understanding of how the surrounding area may influence house price, the application of the regression algorithms to new inputs will likely be limited. That is, the models would only be able to predict new inputs for houses also found in Ames, since the neighborhoods in the training set are all within said city. Possible solutions could be to either eliminate the variable entirely for the sake of simplicity, or to transform it to something more general such as the average wealth of the neighborhood. As for the problem regarding the current economic state, it may be best to avoid the variables entirely and instead focus more on specific aspects of the house. In short, it is reasonable to assume that the economic state of the country can be simply represented through the year in which the house was sold. While it is probable that the price of the house largely depends upon the economic state, the time of the sale will have underlying connotations for the disclosed economic variable. For example, a house sold during a recession would likely have a lower price than a house sold during a period of economic growth. That is, a house may have

sold for more in 2006 than it would have in 2008 during the recession. The following section will discuss related work which has influenced the approach to the problem.

II - Related Work

Many studies and analyses have been conducted in relation to house prices. There are hundreds of potential variables which could be used to explain these prices, so determining which are most important while maintaining some form of generality is a great challenge. The aforementioned studies have sought to achieve this goal, and the pertinence of the topic means there are a great deal of potential studies to review. That said, there are certainly patterns which can be found in said studies, and there are generally agreed upon beliefs about house pricing from both a mathematical and an economic perspective.

From the economic perspective, it is obvious that the supply of houses is crucial in their resultant pricing. If there is a small supply of houses, the demand for them will increase, which in turn drives up the price. For example, 700,000 houses were constructed during the Irish property boom in between 1996 and 2006 [1]. Inevitably, the property industry regressed back to a more stationary point, decreasing demand, leaving an oversupply of houses. With the supply now exceeding demand, prices decreased. One may imagine the inverse scenario, in which demand exceeds supply, thereby increasing prices. According to Opendoor, other economic factors are employment rate and interest rate [2]. If the former is high, then there are more people making money, and they are therefore more likely to want to purchase homes, increasing prices. If the latter is low, then more people are able to afford mortgages for homes, also increasing prices. Further, one might consider the transportation aspects of the house. This could include the proximity of the house to various landmarks such as school and city hubs, and the ease by which these external factors are reached. While all of these factors are undeniably significant in affecting house prices, this study will focus more on the specific features of the house itself.

From the mathematical perspective, it is often found that two of the most significant factors are house quality and area. It can be easily seen through scatter plots that both of these variables have a positive relationship with house prices. That is, the larger and/or the higher quality the house, the more expensive it will be. While this trend is often visualizable, it is crucial that it is supported through statistical analyses, namely significance tests. This is to say that in order for a variable to be considered statistically significant in determining an output, it must have an associated slope parameter which is significantly different to 0. This is one of the primary aspects of regression in statistics. A study conducted in 2021 of the Boston housing dataset [3] combined multiple linear regression with Spearman correlation coefficient analysis in order to determine the most influential factors on house prices. It found that such a combination is an effective means of predicting house prices, albeit with limited universality [4]. Another difficulty the study encountered was the issue of outliers, which led to high residual values and inaccurate predictions for some simulations of test data.

III - Methods

The focus will primarily be on multiple linear regression, along with other regression models which are more advanced. Linear regression will provide a strong initial understanding of the most significant factors affecting house prices, while ridge regression and gradient boosting will enhance this understanding. All algorithms were applied through Python on Google Colab. It should also be noted that each regression technique used in this project is supervised, since the goal is to build a prediction model based on training data.

1. Multiple Linear Regression

- This is a relatively simple algorithm which is used to determine the effect of input variables on a specific response variable. Mathematically, the algorithm is used to learn a function f which takes new data and produces a prediction. The general form of multiple linear regression is included below, with each β corresponding to a model parameter and thus the effect of each explanatory variable x_i , p referring to the number of parameters in the model, and ε referring to random noise for which the predicted model cannot account. It is assumed that the noise is normally distributed with mean 0 and finite variance σ^2 .

$$y = \beta_0 + \beta_1 x_1 + \dots + \beta_p x_p + \varepsilon$$

The algorithm determines the value of each parameter which minimizes errors, resulting in a fitted equation. In turn, it is used to make predictions on new inputs.

$$\hat{f}(x) = \hat{\beta}_0 + \hat{\beta}_1 x_1 + \dots + \hat{\beta}_p x_p$$

- There are five pertinent assumptions that must be made before the implementation of a linear regression algorithm. [5]
 - i. Linear Relationship - There is some sort of linear relationship between the explanatory variables and the response variable. This linearity is in reference to the parameters rather than the variables themselves.
 - ii. Multivariate Normality - The data should be normally distributed. According to the central limit theorem, the normality of errors is not needed as sample size increases so long as this assumption holds.
 - iii. Limited Multicollinearity - There should be as little redundancy between variables as possible. This is explained in more detail in the section on ridge regression.
 - iv. No Autocorrelation - This refers to the relationship between errors. If there is some sort of pattern that can be seen in the residuals produced by the model, there is a high probability that the model is biased and therefore limited in its reliability and usefulness.
 - v. Homoscedasticity - The variance of the errors must be constant. If this assumption is not met, it means that the spread of the residuals are dependent upon factors other than the explanatory variables, leading to skewed and biased predictions.
- The effectiveness of the algorithm will be assessed through the root mean squared error (RMSE) of the residuals. It is calculated by summing the squared difference between each predicted response value and the corresponding actual response values given by the data, divided by the sample size. Finally, the positive square root will be calculated. A lower value for this metric suggests the model fits the data relatively well, while a higher value suggests it does not.

$$RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (\hat{f}(x_i) - y_i)^2}$$

In addition to this, the adjusted R^2 value will be considered. It refers to the proportion of the variance in the response which is explained by the model. Higher values mean the model explains more of the variance observed in house

prices. This value should be adjusted according to the number of parameters in the model, since R^2 naturally increases as the number of parameters increases.

$adj. R^2 = 1 - \frac{\text{Sum of square residuals} / (n-K)}{\text{Sum of square total} / (n-1)}$, with n: sample size, K: parameters

- The output of the linear regression algorithm gives p-values for each parameter in the regression. Small p-values suggest the explanatory variable corresponding to the parameter is statistically significant in explaining the response variable. This fact will be used to perform model selection. That is, variables deemed to be insignificant will be removed from the regression so that the equation focuses only on the most important variables. This will increase the utility of the model.

2. Ridge Regression

- Ridge regression techniques are primarily used on datasets with a considerable amount of multicollinearity. Multicollinearity happens when at least one explanatory variable in the algorithm can be represented by one or more of the other explanatory variables. That is, there is at least one variable which is redundant of the others. It is crucial to reduce this as high multicollinearity between variables reduces the reliability of statistical inferences because it increases the variance of predictions, leading to highly inaccurate predicted values when compared to actual values.
- This algorithm uses l_2 regularization, and seeks to minimize the following loss function. w represents the weights of the algorithm, while i represents the observation of the data set. λ is some constant which scales the magnitude that is the l_2 regularization. [6]

$$\min \left(\sum_{i=1}^n ((x^{(i)})^T w - y^{(i)})^2 + \lambda \|w\|^2 \right)$$

- The first step of the algorithm is to standardize both the explanatory and dependent variables by subtracting their respective means then dividing by their respective standard deviations. Though the variables are adjusted back to their original values once regression coefficients are obtained, the calculations of the regression are strictly based upon standardized variables.
- One of the most important aspects of this algorithm is to balance the bias and variance tradeoff. As the value of λ increases, the bias is expected to increase. If the value of λ were to decrease, the variance would increase. The general trend is that λ is directly related to bias, but inversely related to variance. This means that finding the ideal value of the parameter is challenging since it is complicated to minimize bias and variance simultaneously. [7]
- Aside from the implementation of l_2 regularization, ridge regression is almost identical to linear regression and has the same assumptions of linearity, constant variance, and independence.
- The ridge regression algorithm, due to its production of a fitted equation, will use the same assessment metrics as the multiple linear regression model. Using the same metrics and loss functions as the other algorithms will also allow their performances to be more easily compared.

3. Gradient Boost Regression

- In general, gradient boosting is an algorithm which combines several simpler prediction models into an ensemble. In machine learning, an ensemble is created when learning algorithms are merged together to create stronger predictions than any of the individual models would make on their own. That is, the idea of ensembling is to produce a model which is greater than the sum of its parts. Ideally, such a model would be the best performing on a given dataset and produce the smallest error totals.
- This regression model is an iterative procedure in which the model improves itself at each step by adding an additional estimator $h_m(x)$, where m represented the current iteration or stage. If the model is represented by F_m , this procedure can be denoted by the following for each data point i .

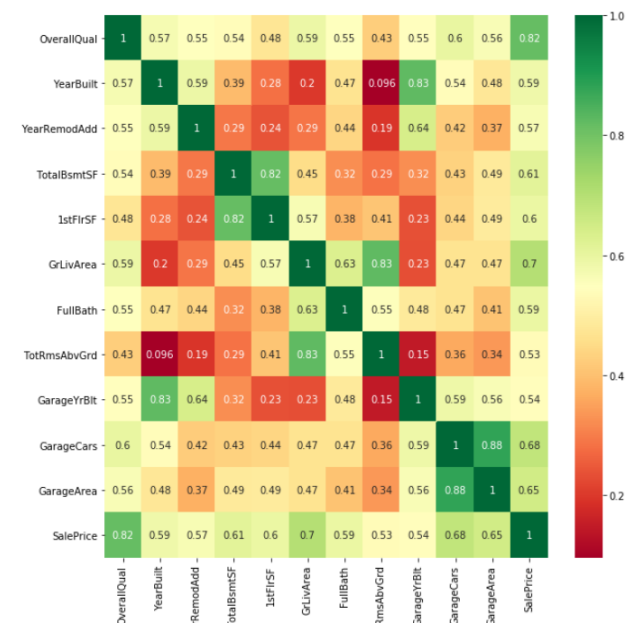
$$F_{m+1}(x_i) = F_m(x_i) + h_m(x_i) = y_i$$
, or

$$h_m(x_i) = y_i - F_m(x_i)$$
- The latter equation gives the residual of the predicted value. This is to say that the gradient boosting algorithm fits h_m to the residual of each prediction. It could be said that the corrections that occur are the “boosts” of the algorithm. The weak learners (h_m) by which the model is constructed are decision trees. These trees are models of decisions and their corresponding outcomes, and can include probabilities, cost of resources, and utility. The trees are often constrained in terms of layers in order to reduce the problem of overfitting. If the tree is too large, that is it contains a great number of branches and therefore encompasses a wide range of outcomes, it is more likely to fit the data extremely closely, which can in turn reduce the reliability of predictions.
- More formally, the algorithm seeks to minimize a loss function $L(y, F(x))$, often through the use of gradient descent. As trees are added to the model, gradient descent is utilized to minimize the weights of each iterative step. The trees are parametrized, and gradient descent moves the loss function downward [8]. The reason gradient descent is used over other optimization methods is that it is guaranteed to move downward. This is because steps are repeatedly taken in the opposite direction of the gradient of the function, which is the direction of steepest descent.
- Decision trees are added and loss is minimized with gradient descent until the algorithm reaches a satisfactory loss. That is, the algorithm will ideally converge so long as enough trees are added to the model.
- This algorithm will also use root mean squared error and adjusted R^2 as its assessment metrics, since it too provides a fitted equation that can be used for predictions.
- It is worth mentioning that there are a number of improvements which can be applied to gradient boosting, they are as follows.
 - i. Tree Constraints - As mentioned before, including too many trees or trees which have many layers can lead to overfitting. As such, limitations can be imposed upon the size and shape of trees to combat this issue. Balancing accuracy while reducing overfitting is a great challenge.

- ii. **Weighted Updates** - The algorithm can be improved by applying respective weights to each tree. Some decision trees may be more efficient than others, so it is logical that it carries more weight than those that are less efficient or less accurate.
- iii. **Random Sampling** - Also known as a stochastic random forest process, this can reduce the correlation between trees that are added to the algorithm. This involves sub-sampling the data, and can be compared to a method of reducing multicollinearity or redundancy between trees. Generally, a stochastic process is meant to create a more efficient and more interpretable model by limiting said redundancy.
- iv. **Penalized Learning** - This involves regularization, often l_1 or l_2 , and considers the complexity of the model based upon the number of leaves (the lowest layer on each decision tree). The idea is to remove branches which do not reduce the loss by an acceptable level, thereby reducing the risk of overfitting.

IV - Experiments

The dataset selected for this project has more than 75 explanatory variables, so there will certainly be a large amount of missing data [9]. Additionally, the large number of explanatory variables could make future predictions more challenging because of the amount of information needed. To combat this, the variables with the least variance in values and those with the most missing observations will be removed from the regressions. Specifically, variables with more than 50% of data missing will be removed, and highly skewed data will **be transformed logarithmically so that each variable is on a comparative scale.** Most of the variables correspond to specific details about the house. These include but are not limited to: square footage (including variables for each story of multi-story homes), number of bedrooms/bathrooms, time of construction, **features such as fireplaces/garages/decks/porches/fences/pools, and qualitative variables indicating the condition of several aspects.** All houses in the dataset were observed in Ames, Iowa between 2006 and 2010.



The first step is to conduct exploratory data analysis. That is, analyzing any underlying trends in the data which could provide an initial understanding of the most influential variables on house prices. One method of doing so is to calculate the correlation between each variable in the data set. A truncated table including several key variables is included to the left. Cells closer to the green colors indicate a high correlation between variables, meaning they have a strong statistical relationship. Using this table, it can be seen that overall quality, garage living area, and the number of cars which can fit in the garage have the highest correlation with sale price.

Before implementing any algorithms, it could be argued that these three variables are the most influential in determining house prices. Therefore, an efficient model for predicting price may limit itself to only these variables. The table can also be used to address the issue of multicollinearity. If the correlation coefficient between any two explanatory variables is deemed too large, it may be wise to remove one or both of the variables from the regressions. This decision can also be influenced by the calculation of variance inflation factors (VIFs). However, the primary objective of this study is to give the most accurate house price predictions possible, with less regard for optimizing the efficiency of the model. That said, there will be some level of model selection such that interpretation remains feasible.

The table below includes the loss functions for each of the algorithms used on the data set [10].

Model	RMSE (Root Mean Squared Error)	SD (Standard Deviation)
Linear Regression		
Ridge Regression	1.1464170875943087	0.0206914845254917
Gradient Boost Regression	1.1341844040912687	0.0314504703195094

1. Multiple Linear Regression Discussion
*** address over/underfitting (model/order selection)
2. Ridge Regression Discussion
3. Gradient Boost Regression Discussion

V - Conclusion

To be completed

References

- [1] T. Pettinger. “Factors that affect the housing market” Economics Help.
<https://www.economicshelp.org/blog/377/housing/factors-that-affect-the-housing-market/>
[Accessed Nov. 15, 2022]
- [2] J. Gomez. “8 Critical Factors that Influence a Home’s Value” Opendoor.
<https://www.opendoor.com/w/blog/factors-that-influence-home-value>
[Accessed Nov. 15, 2022]
- [3] Scikit-Learn. Boston Housing Data Set
https://scikit-learn.org/stable/modules/generated/sklearn.datasets.load_boston.html
[Accessed Nov. 15, 2022]
- [4] Q. Zhang. “Housing Price Prediction Based on Multiple Linear Regression” Hindawi.
<https://www.hindawi.com/journals/sp/2021/7678931/>
[Accessed Nov. 15, 2022]
- [5] “Assumptions of Linear Regression” Complete Dissertation.
<https://www.statisticssolutions.com/assumptions-of-linear-regression/>
[Accessed Nov. 22, 2022]
- [6] P. Ashok. “What is Ridge Regression?” Great Learning.
<https://www.mygreatlearning.com/blog/what-is-ridge-regression/>
[Accessed Nov. 22, 2022]
- [7] S. Fortmann-Roe. “Understanding the Bias-Variance Tradeoff”
<http://scott.fortmann-roe.com/docs/BiasVariance.html>
[Accessed Nov. 21, 2022]
- [8] J. Brownlee. “A Gentle Introduction to the Gradient Boosting Algorithm for Machine Learning” Machine Learning Mastery.
<https://machinelearningmastery.com/gentle-introduction-gradient-boosting-algorithm/>
[Accessed Nov. 22, 2022]
- [9] D. De Cock “Ames, Iowa: Alternative to the Boston Housing Data as an End of Semester Regression Project” Journal of Statistics Education, Volume 19, Number 3.
<http://jse.amstat.org/v19n3/decock.pdf>
[Accessed Nov. 17, 2022]
- [10] J. Park “House Prices Project Final” Google Colaboratory.
<https://colab.research.google.com/drive/1CnfmipFwtsTrs9UiDfSsT>
[Accessed Nov. 25, 2022]